August 5, 2026

Dear Hiring Team,

I am writing to apply for the Forward Deployed Machine Learning Engineer position with Stratum AI. With a Master of Science in Computer Science from the University of Calgary and applied experience in Python, PyTorch, CNN-based modelling, geospatial data processing, and research software, I am excited by the opportunity to help build machine learning systems that support real mining decisions.

My graduate research focused on developing machine learning and data-management systems for large geospatial and Earth observation datasets. In my digital elevation model super-resolution project, I designed Python and PyTorch workflows for data collection, preprocessing, model training, and evaluation. I trained deep ResNet models for 4x terrain reconstruction and compared model quality using RMSE, slope, TRI, TPI, and wavelet-based metrics. This work required careful experimentation, debugging, and practical judgment about whether model results generalized beyond convenient random splits.

I also developed C++ and Python software for scalable processing, storage, and retrieval of satellite imagery through my work with BigGeo, Mitacs, and the University of Calgary. I built DGGS-based encoding pipelines, designed tensor-based storage for multiresolution retrieval, and improved processing performance with C++ multithreading. These projects strengthened the engineering habits I would bring to Stratum AI: writing modular code, working with complex spatial datasets, validating performance, documenting technical decisions, and making research outputs usable by others.

The forward-deployed aspect of this role is especially interesting to me because it connects model development with real operational context. As a teaching assistant and deep learning mentor, I have often translated technical methods for different audiences, explained model behavior and data-processing choices, and helped others move from ambiguous problems toward clear next steps. I would be glad to bring that communication style to work with engineers, geologists, and client teams while continuing to grow in mining-specific resource modelling and deployed ML systems.

Stratum AI stands out to me because the role combines PyTorch, applied deep learning, large scientific datasets, model evaluation, and direct stakeholder impact. My background is not from mining, but it is strongly aligned with geospatial data, scientific imaging, deep learning experimentation, and practical research engineering. I am confident I can contribute quickly while learning the domain with focus and humility.

Thank you for considering my application. I would welcome the opportunity to discuss how my machine learning research, software engineering experience, and interest in real-world spatial modelling can contribute to Stratum AI.

Sincerely,
Amir Mirzai Golpayegani
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